Amitaditya Paul

Clear Roads, Clearer Skies: Assessing the Environmental Impact of Immediate Roadside Prohibition in British Columbia

Research Question and Objectives: This study aims to answer the following question, "What is the effect of vehicle traffic on air quality in British Columbia?" The two objectives of this study are, 1) to assess the impact of changes in traffic patterns, due to altered B.C. laws on intoxicated driving, on key air quality variables; and 2) to conduct a cost-benefit analysis (CBA) evaluating the policy's impact on vehicle traffic and air pollution.

Context: On September 20, 2010, the B.C. provincial government introduced the Immediate Roadside Prohibition (IRP) program to instantly remove alcohol-affected drivers from the roads, resulting in a sustained reduction in alcohol-related motor vehicle fatalities [1]. While studies indicate a reduction in vehicle traffic among alcohol consumers [2], the broader impact of the IRP on vehicle traffic patterns and subsequent air quality remains unexplored. Research in urban traffic management suggests a correlation between vehicle flow and air pollution [3], yet the specific influence of traffic law enforcement strategies like the IRP on this dynamic is not well-documented. This study aims to fill this gap by examining how changes in vehicle traffic, influenced by the IRP program, affect urban air quality in British Columbia.

Theory and Methodology: The analysis will work by estimating an instrumental variables (IV) design, using IRP as an instrument for vehicle traffic. This requires that the introduction of IRP is correlated with vehicle traffic but does not affect air quality except through traffic. We estimate the model using two-stage least squares. In the first stage, Ordinary Least Squares (OLS) regression is used to predict vehicle traffic based on the instrumental variable. This step isolates the variation in traffic that is solely attributable to the policy change. Then, in the second stage, another OLS regression is conducted, using the predicted traffic values from the first stage as the independent variable, to estimate the effect on air quality. It is crucial to perform an F-test on the coefficient of IRP in the first stage regression to verify that the F-statistic is greater than 10, ensuring a strong instrument [4]. The hypothesis is the introduction of the IRP program will result in a reduction in vehicle traffic and air pollution i.e., negative, and positive coefficients respectively. Data needed for the regressions are from the B.C. government websites [5], [6].

Next, we will quantify the economic costs and benefits, and economic implications of the IRP introduction, focusing on changes in vehicle traffic and air pollution. Following the hypothesis, studies show that key benefits include fee revenues, improved road safety, reduced social costs of collision from fewer alcohol-impaired driving incidents, decreased traffic congestion leading to economic gains from improved traffic flow, and enhanced air quality resulting in lower medical expenses. Costs could include policy enforcement costs, loss in tax revenues, and the societal burden of adapting to the new policy for both alcohol consumers and non-consumers [7], [8]. The net economic benefit will be determined as the Net Present Value (NPV), calculated by discounted benefits less discounted costs. The hypothesis is that the program will result in a positive NPV. Data sources for the CBA include government articles and websites [1], [9], [10], [11], and articles from an independent research organization [8], [12].

Significance: This study provides insights on how traffic policies like the IRP can enhance road safety and air quality, which has implications for both Canada and globally, particularly in refining existing alcohol-impaired driving laws [13]. Its CBA framework also aids in understanding the monetary aspects of environmental policies, thus informing sustainable initiatives in an environmentally conscious world.

Fit and Interest: During my undergraduate studies in B.C., I delved into the practical analysis of diverse B.C. projects and policies, which has significantly shaped my approach. This study aligns well with my background, given my extensive experience in analyzing policy effects as a Research Assistant.

Bibliography

[1] Ministry of Public Health and Solicitor General. 2023. “Immediate Roadside Prohibition Review Policies.” Accessed November 17, 2023. https://www2.gov.bc.ca/assets/gov/driving-and-transportation/driving/roadsafetybc/publications/irp-review-policies_updated_jan_2023.pdf.

[2] Beirness, Douglas J., and Erin E. Beasley. 2014. “An Evaluation of Immediate Roadside Prohibitions for Drinking Drivers in British Columbia: Findings from Roadside Surveys.” *Traffic Injury Prevention* 15 (3): 228–233. https://doi.org/10.1080/15389588.2013.813628.

[3] Xia, Liping, and Yaping Shao. 2005. “Modelling of Traffic Flow and Air Pollution Emission with Application to Hong Kong Island.” *Environmental Modelling & Software: With Environment Data News* 20 (9): 1175–1188. https://doi.org/10.1016/j.envsoft.2004.08.003.

[4] Andrews, Isaiah, James H. Stock, and Liyang Sun. 2019. “Weak Instruments in Instrumental Variables Regression: Theory and Practice.” *Annual Review of Economics* 11 (1): 727–753. https://ssrn.com/abstract=3445885.

[5] Government of British Columbia. N.d. “Traffic Data Program.” Accessed November 17, 2023. https://www.th.gov.bc.ca/trafficData/index.html.

[6] Government of British Columbia. N.d. “Welcome to the BC Air Data Archive Website.” Accessed November 17, 2023. https://envistaweb.env.gov.bc.ca.

[7] Kenkel, Donald. 1998. “A Guide to Cost-Benefit Analysis of Drunk-Driving Policies.” *Journal of Drug Issues* 28 (3): 795–812. https://doi.org/10.1177/002204269802800313.

[8] Victoria Transport Policy Institute. 2020. “Transportation Cost and Benefit Analysis II – Air Pollution Costs.” Accessed November 17, 2023. https://tc.canada.ca/en/road-transportation/statistics-data.

[9] Transport Canada. 2023. “Statistics and data.” Accessed November 17, 2023. https://tc.canada.ca/en/road-transportation/statistics-data.

[10] BC Ministry of Transportation and Infrastructure. 2018. “Default Values for Benefit Cost Analysis In British Columbia 2018.” Accessed November 17, 2023. https://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/planning/tools/default_values-benefit_cost_analysis.pdf.

[11] Government of British Columbia. 2011 “Budget and Fiscal Plan 2011/12 – 2013/14.” Accessed November 17, 2023. https://www.bcbudget.gov.bc.ca/2011/bfp/2011_budget_fiscal_plan.pdf.

[12] Victoria Transport Policy Institute. 2016. “Transportation Cost and Benefit Analysis: Techniques, Costs and Implications.” Accessed November 17, 2023. https://tc.canada.ca/en/road-transportation/statistics-data.

[13] R. Solomon, and E. Chamberlain. 2014. “Federal Impaired Driving Policy: Moving Beyond Half Measures.” *Canadian Public Policy/ Analyse de Politiques* 40 (1): 15–30. https://www.jstor.org/stable/24365074.